Session setup - do it each time you log into Ares during this training:

```
source $PLG_GROUPS_STORAGE/../tutorial/2026X/RunningEEBG/source_me.sh
cd $MYDIR
ls
```

! it's really important to source and not run the script ~~./source_me.sh~~

You can now use `cd $MYDIR` to go back to get to `$SCRATCH/szkolenia-cyfronet/2026X/RunningEEBG`
Also an alias for `mc` has been set, so the theme is "dark" now.

! *blue text* is commentary

! [] square brackets - use a specific key on your keyboard like [Esc], [Tab] or [Up] (up arrow key)

! <> brackets - to put in their place a specific phrase, usually output of a previous command

! VIEW EVERY BATCH SCRIPT BEFORE SENDING IT TO THE QUEUE

- mc -e script.slurm ; *view the script* ; [Esc][Esc]
- diff basic.slurm script.slurm ; *compare the scripts*

Batch scripts overview

! start in $MYDIR/Exercises/Run

1. sbatch basic.slurm ; squeue --me ;
   cat slurm-*.out ; cat basic.log ;
   tail -8 basic.log

2. sbatch overview.slurm ; *wait 1 minute* ;
   cat overview.err ; *read the overview* ;
   cd $SCRATCH/slurm_jobdir/<jobID> ; ls ;
   cd $MYDIR/Exercises/Run

3. sbatch tasks.slurm ; hpc-jobs ;
   sbatch taskspnode.slurm ; hpc-jobs ;
   compare Cores and Nodes columns - no difference

4. sbatch memory.slurm ; hpc-jobs ;
   sbatch memorypcore.slurm ; hpc-jobs ;
   compare Decl. mem column - no difference

5. sbatch gpu.slurm ; hpc-jobs ;
   look at GPUs column

6. sbatch envars.slurm ; cat envars.out

Modules

! start in **$MYDIR/Exercises/Modules**

1. module load AlphaFold ; module list ; ml list ; ml

2. ml unload AlphaFold ; ml VSCode ; ml unload VSCode ; ml ; ml purge ; ml

3. module load python ; *read the error* ; ml spider python ; ml avail python ; *compare* ; ml spider Python/3.11.5 ; *lookup the toolchain to load* ; ml GCCcore/13.2.0 Python/3.11.5 ; ml purge

4. hpc-modules -s python ; hpc-modules -s gromacs ; ml GCC/14.2.0 OpenMPI/5.0.7 GROMACS/2026.2-CUDA-12.8.0 ; ml purge

5. sbatch modules.slurm ; cat *.out ; cat *.err ; ml GCCcore/14.2.0 Python/3.13.1 ; sbatch modules.slurm ; *compare .err files* ; ml purge

6. *On Helios if you have access* ; hpc-modules -s python ; *read the error* ; hpc-modules x86 -s python ; hpc-modules gh200 -s python ;

Job Monitoring

! start in **$MYDIR/Exercises/Monitor**

1. hpc-[Tab][Tab] ; *lists all the commands* ;
   ls /opt/cyfronet/bin -1 | grep "hpc-"

2. hpc-fs ; echo $HOME ; echo $SCRATCH ;
   echo $PLG_GROUPS_SOFTWARE ;

3. hpc-grants ; *see all grants, but not on tutorial accounts, Trainer uses:*
   hpc-grants | grep -A 18 "Grant: plgtraining2026"

4. sbatch correct.slurm ; squeue --me ;
   copy the jobID ; scancel <jobid> ;
   squeue --me ; cat correct.err ;
   tail -1 correct.err

5. hpc-jobs-history ; *view your past jobs from today* ;
   Trainer is using this command instead:
   hpc-jobs-history | head -1 ; hpc-jobs-history | tail -5

6. sbatch timelimit.slurm ; *wait 1 min* ;
   grep "DUE TO TIME LIMIT" timelimit.out ;
   tail -30 timelimit.out

7. sbatch nogpu.slurm ; cat gpu.out ;
   grep -i "no gpu" nogpu.out

8. sbatch oom.slurm ; cat oom.out ;
   grep -i "out of memory" oom.out ; *nothing* ;
   grep -i "oom" oom.out

9. sbatch missingfile.slurm ;
   tail -30 missingfile.out ;
   grep "file" missingfile.out

10. sbatch wrongsyntax.slurm ; cat wrongsyntax.out ;
   gromacs accepts 0 tasks as all available ;
   diff correct.slurm wrongsyntax.slurm ;
   diff wrongsyntax.slurm wrongsyntax2.slurm ;
   sbatch wrongsyntax2.slurm ; cat wrongsyntax2.out

11. grep -i "error" *.out

12. sbatch connect.slurm ; ssh_slurm <jobid> ;
   copy the node ; ssh_slurm <jobid> <node> ; htop ;
   observe how processes run ; [F10] ;
   echo "SCRATCHDIR: $SCRATCHDIR" ;
   echo "SLURM_SUBMIT_DIR: $SLURM_SUBMIT_DIR" ; echo
   "SLURM_JOB_ID: $SLURM_JOB_ID" ;
   echo "SLURM_MEM_PER_CPU: $SLURM_MEM_PER_CPU" ;
   echo "SLURM_NTASKS: $SLURM_NTASKS" ;
   compare with envars.out ;
   Ctrl+C then Ctrl+D to return to the login node ;
   scancel <jobid> ; *to not clog up the queue*

! this job runs about 5 min, if time runs out, start over, but first make sure you are on the login node, your should see in the prompt: [ares][plguser@**login01** Monitor] otherwise Ctrl+C then Ctrl+D to return to login node

HANDS-ON

! start in **$MYDIR/Exercises/HANDS-ON**

1. modules: Choose any one module of GROMACS that:
   - was built using any version of MPI,
   - whose version starts with 2024.

   Add a line that will load that module to addmodule.slurm. Check your .err file for errors, when the calculation runs correctly, the exercise is finished.

   ! HPC Handout lists possible tools to browse available module versions

2. scalability: Change the gromacs.slum scripts in each directory according to instructions below. Check with hpc-jobs or hpc-jobs-history if your allocation was correct.
   - for *n* in 2/ 4/ 8/ 16/ - run with *n* number of tasks per node
   - gpu-vs-cpu/ - run cpu/ and gpu/ without changes

   Once all are finished, run the ./result-calc.sh script to generate the results. You can save them using ./result-calc.sh > results.dat

3. iwontruns: Go through each of four iwontrun_*.slurm scripts and diagnose what is causing their problems.

   ! there might be more than one problem per script

4. copylogs: At the end of the copylogs.slurm, add commands that will move the md.log file to the directory where the job has been started (where .out and .err files appear).

5. broken: Determine what is wrong and fix the batch script broken.slurm. There is more than one thing wrong. Do not use more than 1 core. The exercise is complete, when the md.log contains the time and performance information at the end.

   ! if you have problems, look into the $MYDIR/Justincase/Solutions for expected outputs and hints
   the solution is also presented, but try yourself first